

2. Matthew Portnoym, *Virtualization Essentials*, John Wiley and Sons Publication, 2012
3. Thomas Erl, *Service oriented Architecture*, Pearson publications, 2016
4. Scott D Lowe, *Hyper converged Infrastructure implementation strategies*, Actual Tech media, 2015
5. George Reese, *Cloud application architectures: building applications and infrastructure in the cloud*, O'Reilly Media, Inc., 2009.
6. Kai Hwang, Geoffrey Fox, Jack Dongarra, Todd Green, *Distributed and Cloud Computing: Clusters, Grids, Clouds and The Future Internet*, Morgan Kaufmann Publishers, 2011

Reference Books:

1. Rajkumar Buyya, James Broberg, Andrzej Goscinski, *Mastering Cloud Computing*, McGraw Hill Education (India) Private Limited New Delhi, 2013.
2. Gautam Shroff, *Enterprise Cloud Computing*, Cambridge University Press, 2012.
3. John Rhoton, *Cloud Computing Explained*, (2e), Recursive Press, 2010. Barrie Sosinsky, *Cloud Computing: Bible*, Wiley India, 2011

DEEP LEARNING [Revised Credit System] (Effective from the academic year 2022 onwards) SEMESTER – VI/VII			
Subject Code	CSE_4064	IA Marks	50
Number of Lecture Hours/Week	03	Exam Marks	50
Total Number of Lecture Hours	36	Exam Hours	03
CREDITS - 03			
Course objectives: This course will enable students to - To introduce the mathematical tools for neural networks - To provide exposure to the concepts of deep learning - To provide exposure to TensorFlow for deep learning			
Module -1			Teaching Hours
INTRODUCTION AND PRELIMINARIES: Why deep learning? Trends in Deep Learning Mathematical Preliminaries: Linear Algebra, Probability and Information Theory, Numerical computation Machine Learning Basics: Learning Algorithms, Capacity, Under and Overfitting, Hyperparameter and Validation Set, Estimators, Bias and Variance, MLE, Bayesian Statistics, Supervised Learning Algorithms, Unsupervised Learning Algorithms			07 Hours
Module -2			

Deep Feedforward Networks: Learning XOR, Gradient-Based Learning, Hidden Units, Architecture Design, Back-Propagation Algorithm	05 Hours
Module - 3	
Regularization for Deep Learning: Parameter Norm Penalties, Norm Penalties as Constrained Optimization, Regularization and Under-Constrained Problems, Dataset Augmentation, Noise-Robustness, Semi-Supervised Learning, Multi-Task Learning, Early Stopping, Parameter Tying and Parameter Sharing, Sparse Representations, Bagging and Other Ensemble Methods, Dropout, Adversarial Training	06 Hours
Module-4	
Optimization for Training Deep Models: Challenges in Neural Network Optimization, Basic Algorithms, Parameter Initialization Strategies, Algorithms with Adaptive Learning Rates, Approximate Second-Order Methods, Optimization Strategies and Meta-Algorithms	05 Hours
Module-5	
Convolutional Networks: Convolution Operation, Pooling, Convolution and Pooling, Variants of Convolution Function, Structured Outputs, Data Types, Efficient Convolution Algorithms, Random or Unsupervised Features	05 Hours
Module-6	
Recurrent and Recursive Networks: Unfolding Computational Graphs, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to-Sequence Architecture, Deep Recurrent Networks, Recursive Neural Networks, Echo State Networks, LSTM Practical Methodology: Performance Metrics, Default Baseline Models, Selecting Hyperparameters, Debugging Strategies	08 Hours
Course outcomes:	
After studying this course, students will be able to: 1. Understand different neural network architectures. 2. Choose right deep learning model and architecture for a given learning problem 3. Understand the necessary parameters in the design of deep networks. 4. Implement deep learning architecture on platforms like Caffe, Theano or TensorFlow 5. Comprehend and communicate the content of a research paper in the area of deep learning	
Text Books:	

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press 2016.
2. Simon Haykin, *Neural Networks and Learning Machines*, PHI, 2008
3. Andrew Ng's *Notes on Machine Learning* from CS229.

HUMAN COMPUTER INTERFACE [Revised Credit System] (Effective from the academic year 2022 onwards) SEMESTER – VI/VII			
Subject Code	CSE_ 4069	IA Marks	50
Number of Lecture Hours/Week	03	Exam Marks	50
Total Number of Lecture Hours	36	Exam Hours	03
CREDITS - 03			
Course objectives: This course will enable students to 1. To stress the importance of a good interface design. 2. To understand the importance of human psychology in designing good interfaces. 3. To motivate students to apply Human Computer Interface in their day – to – day activities. 4. To encourage students to indulge into research in Machine Interface Design.			
Module -1			Teaching Hours
THE HUMAN: Introduction, Input–output channels, Getting noticed, Design Focus: Where's the middle, Human memory, Design Focus: Cashing in, Thinking: reasoning and problem solving, Human error and false memories, Emotion, Individual differences, Psychology and the design of interactive systems. THE COMPUTER: Introduction, Text entry devices, Numeric keypads, Positioning, pointing and drawing, Display devices, Hermes: a situated display, Devices for virtual reality and 3D interaction, Physical controls, sensors and special devices, Feeling the road, Design Focus: Smart-Its – making using sensors easy, Paper: printing and scanning, Readability of text, Memory, Processing and networks, The myth of the infinitely fast machine Text Book 1: Chapter 1 and 2			6 Hours